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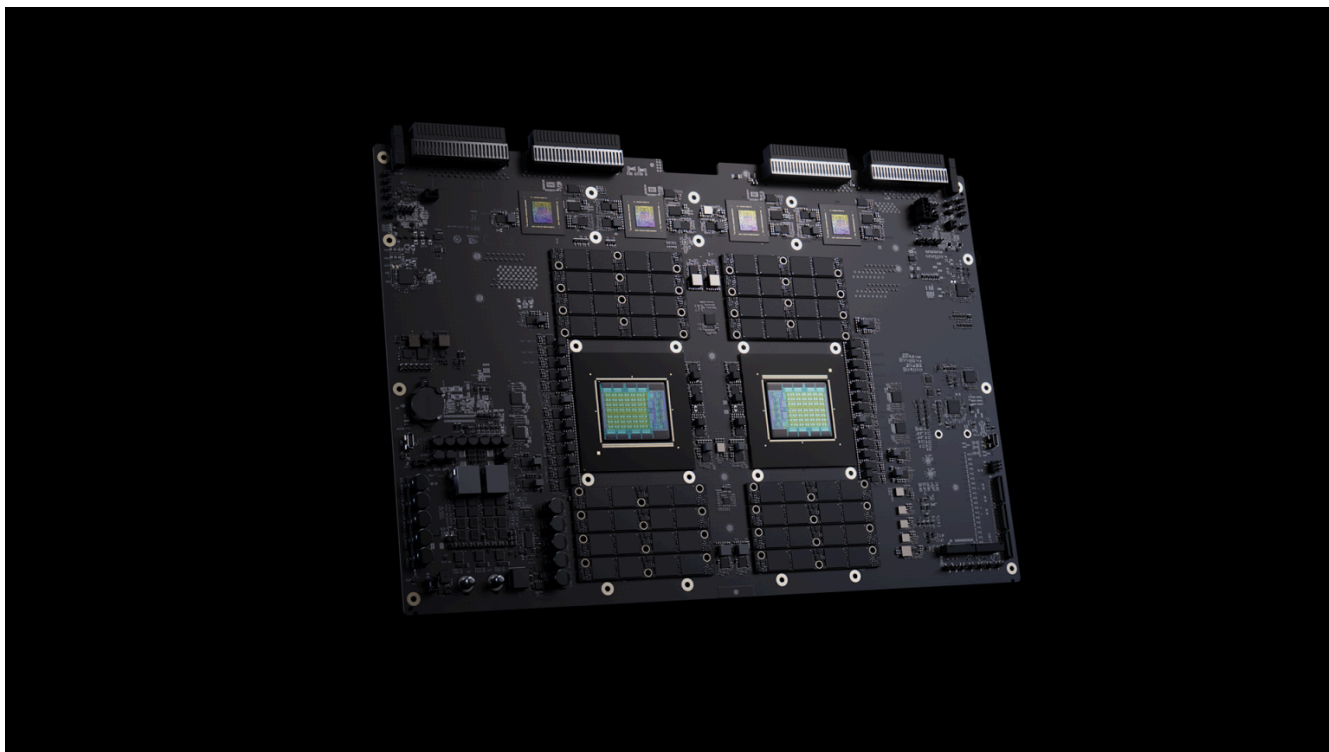
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NVIDIA Vera Storage Benchmarks: более быстрое шифрование, сжатие, проверка целостности и восстановление данных для систем хранения, оптимизированных для искусственного интеллекта



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повторно используют данные из кэша «ключ-значение» (KV), запускают инструменты и генерируют новые результаты, системы хранения данных должны непрерывно предоставлять и сохранять информацию, необходимую для работы цикла рассуждений агента.

Каждый шаг агента может инициировать несколько операций хранения данных, и эти операции могут повторяться тысячами одновременно работающих агентов со все более широкими контекстными окнами. Для предоставления и сохранения этих данных требуется нечто большее, чем простое чтение и запись. Вычисления ИИ выполняются на графических процессорах, но агентские процессы, вызовы инструментов, задачи управления данными и поддерживающие их сервисы хранения данных выполняются на центральных процессорах.

При записи хранилище может сжимать и шифровать данные, вычислять контрольные суммы и определять избыточность. При чтении оно может проверять, расшифровывать, распаковывать или восстанавливать данные перед возвратом в приложение. Эти функции необходимы для обеспечения безопасности и отказоустойчивости систем искусственного интеллекта. Каждая функция также требует дополнительной обработки на центральном процессоре по мере прохождения данных через хранилище.

По мере роста параллелизма агентов (когда несколько пользователей, ИИ-агентов или задач выполняются одновременно) и объемов контекста хранилище должно выполнять все больше этой работы, не ограничивая скорость отклика приложений или генерации токенов. Оно должно предоставлять данные с той скоростью, которая необходима для ускоренных вычислений.

Многие из этих функций находятся непосредственно на пути следования данных; одна отложенная операция может замедлить весь поток данных. Для их масштабирования на обычных процессорах может потребоваться больше ядер, энергии и систем охлаждения, что увеличит стоимость инфраструктуры, но при этом производительность будет зависеть от самого медленного этапа. Более быстрые твердотельные накопители и сети не смогут полностью раскрыть свой потенциал, если процессор, отвечающий за безопасность, защиту и подготовку данных, не будет успевать за ними.

Устранение пробелов в хранении и обработке данных

Процессор хранения данных NVIDIA Vera BlueField-4 STX, ключевой компонент платформы NVIDIA STX для платформ обработки данных на основе искусственного интеллекта,



обеспечения работы графических процессоров NVIDIA Rubin, также ускоряет обработку данных на стороне процессора.

Результаты тестирования показывают, что Vera превосходит процессоры x86 по скорости шифрования и дешифрования, восстановления, проверки целостности, сжатия и распаковки, а также по производительности многоступенчатого конвейера хранения данных. Благодаря этому платформы хранения данных могут обрабатывать больше данных и применять основные корпоративные сервисы с меньшими затратами ресурсов процессора и энергии, а более высокая скорость сжатия помогает снизить требования к объему хранилища и пропускной способности.

В этом посте рассказывается о том, как процессор Vera в BlueField-4 STX ускоряет обработку данных, необходимую для агентного ИИ, помогая платформам хранения данных на основе ИИ обеспечивать безопасность, защиту, проверку и сжатие большего объема данных, а также повышать пропускную способность и эффективность обработки данных.

Архитектура процессора Vera: создана для удовлетворения двойных потребностей систем хранения данных

Процессор Vera включает в себя 88 процессорных ядер Olympus, разработанных NVIDIA, которые полностью совместимы с набором инструкций Armv9.2. Процессор поддерживает 176 потоков пространственной многопоточности NVIDIA. Эти ядра работают в паре с масштабируемой когерентной структурой NVIDIA (Scalable Coherency Fabric, SCF), модулем памяти Small Outline Compression Attached Memory Module (SOCAMM2) LPDDR5X, что обеспечивает высокую производительность в однопоточных вычислениях и высокую пропускную способность процессора в масштабах фабрики искусственного интеллекта.

SCF обеспечивает согласованную передачу данных между ядрами, общим кэшем, контроллерами памяти и устройствами ввода-вывода с пропускной способностью до 3,4 ТБ/с и унифицированным кэшем L3 объемом 164 МБ. Это обеспечивает активным ядрам предсказуемый доступ к общим данным с высокой пропускной способностью по мере масштабирования рабочих нагрузок в процессоре. Подсистема памяти SOCAMM2 LPDDR5X дополняет вычислительную сеть совокупной пропускной способностью до 1,2 ТБ/с или до 14 ГБ/с на ядро, обеспечивая ядра NVIDIA Olympus ресурсами для ресурсоемких и высококонкурентных рабочих нагрузок. Модульная память, которую можно заменить в



данных.

Storage primitives place two distinct demands on a CPU. First, within each data stream, encryption, integrity checking, recovery, compression, and decompression must complete quickly before subsequent storage processing can proceed, making sustained per-core performance important. Second, across the system, these operations run over many concurrent streams and repeatedly move data through caches and memory, making bandwidth and predictable latency equally important.

Vera addresses both requirements. The Olympus core combines wide instruction throughput, advanced branch prediction, deep out-of-order execution, and vector and cryptographic resources to help each core sustain instruction throughput across control-heavy and data-processing code.

NVIDIA Spatial Multithreading, the monolithic compute die, SCF, unified L3 cache, and high-bandwidth SOCAMM2 memory help keep active cores supplied with data while reducing thread-to-thread interference and supporting more predictable data access under load. Together, these capabilities help explain the measured gains across encryption, integrity checking, parity calculations, compression, and the multi-stage storage pipeline.

This enables the BlueField-4 STX Storage Processor to sustain more CPU-side storage processing across concurrent data streams without proportional increases in CPU resources, power, and cooling.

Measuring foundational storage performance

Storage tasks execute repeatedly across storage read, write, and recovery paths. Their throughput and efficiency help determine whether CPU-side processing keeps pace with SSDs and networks or becomes the limiting stage in the data path. The storage primitive microbenchmarks in this post isolate these functions to measure the processor's contribution. They show the CPU performance and headroom available in Vera for building higher-throughput, more efficient storage services.

Each test runs within a single process using data already held in memory. The tests exclude file I/O, disk performance, networking, command startup, and external-device bottlenecks unless otherwise identified. The benchmark set uses common libraries, including OpenSSL, Zstandard,



A purpose-built test framework runs each workload consistently and controls buffer sizes, thread counts, CPU placement, timing, correctness validation, and result collection. The algorithms and many of the software implementations are widely adopted. The test framework applies the same workload definitions and controls across Vera and x86, enabling a consistent processor comparison. The results can be reproduced using the source code, scripts, fixed software versions, configurations, and result files, although the complete benchmark set is not an off-the-shelf public benchmark.

These measurements establish Vera's performance on the storage building blocks that influence secure data movement, resilience, capacity efficiency, and service density. Production storage paths often apply several of these operations to the same data, causing their CPU-processing requirements to accumulate.

When combined in production storage software, the performance of these individual operations contributes to aggregate throughput and CPU efficiency. Higher throughput across individual primitives and the multi-stage pipeline gives storage software more CPU headroom to keep pace with SSDs and networks, support concurrent data flows, and apply essential data services efficiently. End-to-end testing is still required to quantify complete storage-system or GPU-performance outcomes.

Securing more AI data with faster encryption

AI factories process sensitive information, including model assets, enterprise knowledge, agent context, prompts, outputs, and customer data. AES-128 is widely used for data-at-rest and data-in-flight encryption. Encryption sits directly in the write path, where its throughput can determine how much secure data a platform can process each second before encryption becomes a bottleneck. Vera delivers up to **1.43x higher AES-128 encryption throughput** than the x86 CPU used for comparison.



Figure 1. NVIDIA Vera accelerates AES-128 storage encryption

Decryption performs the corresponding operation on the read path and Vera delivers up to **1.29x higher AES-128 decryption throughput** than the x86 CPU used for comparison.

Figure 2. NVIDIA Vera accelerates AES-128 storage decryption

Higher encryption throughput enables storage systems to secure more data without constraining writes, while faster decryption reduces the time required to return protected data to agents, applications, or accelerators. These help storage systems protect and return growing volumes of AI data without consuming an increasing share of the storage-performance budget.



Storage platforms use erasure coding to protect data when drives, nodes, or data fragments become unavailable. Reed-Solomon encoding creates redundancy, while recovery uses that redundancy to reconstruct missing or corrupted data. Encoding typically occurs on the write path. Recovery occurs during rebuild, repair, or a degraded read. Their performance results can differ because these operations use different compute and memory-access patterns.

Vera delivers up to **3.26x higher Reed-Solomon throughput** than the x86 CPU in a recovery workload.

Figure 3. NVIDIA Vera accelerates Reed-Solomon storage recovery

Higher Reed-Solomon throughput enables storage systems to write protected data and reconstruct missing data faster. In selected efficiency measurements, Vera also completes more protection work within the available CPU power envelope, helping shorten rebuilds and reduce contention with normal data services.

Validating data integrity at higher throughput

Data must remain correct as it is moved, stored, and retrieved. Cyclic redundancy checks, or CRCs, create checksums that storage systems use to detect accidental corruption.



missing or corrupted information.

A storage system may calculate CRCs on both the write and read paths, including while copying data between buffers. As data volumes grow, these checks can consume a meaningful share of CPU resources.

Vera delivers up to **3.67x higher CRC32C throughput** than the x86 CPU.

Figure 4. NVIDIA Vera accelerates CRC32C storage integrity checking

Higher CRC32C throughput enables storage systems to validate more data without integrity checking limiting reads or writes. For agentic workloads, this helps return reliable context, persistent memory, and enterprise data with less CPU-side processing delay.

Reducing data footprint with faster compression and decompression

Agentic AI creates growing volumes of context, logs, checkpoints, retrieval data, intermediate outputs, and persistent memory. Compression reduces the storage capacity required for this data and the bandwidth needed to move it. Decompression restores the data when it is read. Compression may run inline or after data is written, depending on the storage architecture, while decompression is typically required when compressed data is read. Compression throughput affects how quickly data can be reduced, and decompression throughput can affect



Vera delivers up to **3.29x higher compression throughput** than the x86 CPU, while sustaining its advantage across the measured thread counts.

Figure 5. NVIDIA Vera accelerates storage compression

The decompression benchmark measures the corresponding operation when compressed data is read. Vera delivers up to **1.72x higher decompression throughput** than the x86 CPU under concurrency, with its advantage increasing as more worker threads run in parallel.

Figure 6. NVIDIA Vera accelerates storage decompression



compression throughput enables storage systems to reduce the amount of data written, stored, and transferred while sustaining CPU-side processing performance.

Higher decompression throughput helps storage systems return decompressed data to agents and applications more quickly. Together, these capabilities lower pressure on storage capacity and bandwidth while enabling each processor to compress and decompress more data as agentic AI workloads grow.

Accelerating a multi-stage storage write path

Storage systems rarely execute data services independently. A secure write path may compress data to reduce its footprint and then encrypt it before it is written. The benchmark set includes a memory-resident pipeline that applies compression followed by encryption to each data buffer. Unlike the preceding benchmarks, which measure individual operations, this test measures total pipeline throughput when two CPU-intensive storage functions execute in sequence.

Vera delivers up to **3.21x higher pipeline throughput** than the x86 CPU used in the benchmark for the two-stage compression and encryption pipeline.

Figure 7. NVIDIA Vera accelerates a multi-stage compression and encryption pipeline.



and protects data. Higher multi-stage pipeline performance enables the storage system to process more data per processor, helping sustain write throughput as agentic AI data volumes grow.

Scaling agent execution and storage with Vera

Agentic AI makes CPU execution and storage processing part of the same AI factory data path.

The CPU runs tools, code, retrieval, analysis, and data-processing steps between model calls. The storage system secures, protects, validates, compresses, and returns the data those steps require. Both must scale within the finite CPU, power, and cooling resources available across the AI factory.

Vera was designed to accelerate CPU-dependent work for the agentic AI era. In NVIDIA Vera Rubin, it serves as the host CPU for NVIDIA GPUs and supports agent execution. Standalone Vera delivers up to 1.8x higher performance per core in agentic tools. In BlueField-4 STX, Vera powers the CPU-side processing used by AI-native storage platforms.

The benchmark results show Vera accelerating encryption and decryption for secure data access, Reed-Solomon recovery for faster reconstruction of missing or corrupted data during storage rebuilds and repairs, CRC32C for high-throughput integrity validation, and compression and decompression to reduce capacity and bandwidth demands and accelerate data retrieval.

By sustaining these functions individually and within a multi-stage write path, Vera helps AI-native storage platforms process and return secure, reliable data with less CPU-side delay. It also supports more concurrent data flows and higher service density without proportional growth in CPU resources, power, and cooling. Across selected workloads, Vera also delivers higher measured performance per watt, enabling more CPU-side storage processing within the available processor power budget.

A common Vera CPU architecture and software toolchain across compute and storage provides a consistent foundation for scaling both agent execution and the supporting data infrastructure.

Learn more about Vera CPUs and BlueField-4 STX



NVIDIA BlueField infrastructure processors accelerate storage, networking, and security for AI factories.

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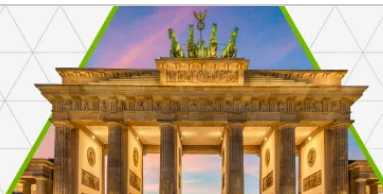
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